```
> restart;#Microwave Planck's low
```

```
> Digits:=25;
```

$$Digits := 25 \tag{1}$$

```
> E(nu):=Em+E0*ln(1+2*Tq*A*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(2\, Tq\, A\, \nu + 1) + C\, \nu^2 \tag{2}$$

```
> Tq:=1.765*10^(-19);mu:=1.25663706212*10^(-6);B:=3.693*10^(-18);
  ec:= 1.602176634*10^(-19);
```

$$Tq := 1.765000000000000000000000 \times 10^{-19}$$

$$\mu := 1.256637062120000000000000 \times 10^{-6}$$

$$B := 3.693000000000000000000000 \times 10^{-18}$$

$$ec := 1.602176634000000000000000 \times 10^{-19} \tag{3}$$

```
> E(nu):=Em+E0*ln(1+A*2.82400000000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{4}$$

```
> A0:=16*Tq;
```

$$A0 := 2.824000000000000000000000 \times 10^{-18} \tag{5}$$

```
> E(nu):=Em+E0*ln(1+A*2.82400000000000000*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := Em + E0 \ln(1 + 2.824000000000000000000000 \times 10^{-18} A\, \nu) + C\, \nu^2 \tag{6}$$

```
> E1:=subs(nu=160.23*10^9,E(nu));
```

$$E1 := Em + E0 \ln(1 + 4.524895200000000000000000 \times 10^{-7} A) + 2.567365290000000000000000 \times 10^{22} C \tag{7}$$

```
> E2:=subs(nu=285*10^9,E(nu));
```

$$E2 := Em + E0 \ln(1 + 8.048400000000000000000000 \times 10^{-7} A) + 8122500000000000000000 C \tag{8}$$

```
> E3:=subs(nu=6.85*10^12,E(nu));
```

$$E3 := Em + E0 \ln(1 + 0.000019344400000000000000000000 A) + 4.692250000000000000000000 \times 10^{25} C \tag{9}$$

```
> solve({E1=7.15,E2=12.64});
```

$$\{A = A,\ C = 1.800136364289894960981063 \times 10^{-23} E0 \ln(1. + 4.524895200000000000000000 \times 10^{-7} A) - 1.800136364289894960981063 \times 10^{-23} E0 \ln(1. + 8.048400000000000000000000 \times 10^{-7} A) + 9.882748639951523335786037 \times 10^{-23},\ E0 = E0,\ Em = 0.462160761894467182056869\, E0 \ln(1. + 8.048400000000000000000000 \times 10^{-7} A) - 1.462160761894467182056869\, E0 \ln(1. + 4.524895200000000000000000 \times 10^{-7} A) + 4.612737417199375170507791\} \tag{10}$$

```
> sol:=solve({E1=7.15,E2=12.64,E3=0},{Em,E0,C,A});
```

$$sol := \left\{A = A,\ C = \left(1.000000000000000000000000 \times 10^{-16}\ \left(715.\ln\left(1.\right.\right.\right.\right.$$

$$\left.\left.\left.\left. + 8.048400000000000000000000 \times 10^{-7}\ A\right) + 549.\ln\left(1.\right.\right.\right.\right.$$

$$\left.\left.\left.\left. + 0.000019344400000000000000000000\ A\right) - 1264.\ln\left(1. + 4.524895200000000000000000\right.\right.\right.\right.$$

$$\left.\left.\left.\left. \times 10^{-7}\ A\right)\right)\right)\Big/\left(5.55513471 \times 10^{8}\ln\left(1. + 0.000019344400000000000000000000\ A\right)\right.$$

$$- 4.68968263471 \times 10^{11}\ln\left(1. + 8.048400000000000000000000 \times 10^{-7}\ A\right)$$

$$+ 4.68412750000 \times 10^{11}\ln\left(1. + 4.524895200000000000000000 \times 10^{-7}\ A\right)\Big),\ E0 =$$

$$-\left(2.578607687773440000000000 \times 10^{12}\right)\Big/\left(5.55513471 \times 10^{8}\ln\left(1.\right.\right.$$

$$\left.\left. + 0.000019344400000000000000000000\ A\right) - 4.68968263471 \times 10^{11}\ln\left(1.\right.\right.$$

$$\left.\left. + 8.048400000000000000000000 \times 10^{-7}\ A\right) + 4.68412750000 \times 10^{11}\ln\left(1.\right.\right.$$

$$\left.\left. + 4.524895200000000000000000 \times 10^{-7}\ A\right)\right),\ Em$$

$$= \left(0.640000000000000000000000\ \left(4.003809021 \times 10^{9}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 0.000019344400000000000000000000\ A\right) - 5.242123046875 \times 10^{12}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 8.048400000000000000000000 \times 10^{-7}\ A\right) + 9.267193750000 \times 10^{12}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 4.524895200000000000000000 \times 10^{-7}\ A\right)\right)\right)\Big/\left(5.55513471 \times 10^{8}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 0.000019344400000000000000000000\ A\right) - 4.68968263471 \times 10^{11}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 8.048400000000000000000000 \times 10^{-7}\ A\right) + 4.68412750000 \times 10^{11}\ln\left(1.\right.\right.\right.$$

$$\left.\left.\left. + 4.524895200000000000000000 \times 10^{-7}\ A\right)\right)\right\}$$

(11)

```
> A:=79904.7*10^1;evalf(sol[4]);#Em
```

$$A := 799047.0$$

$$Em = -1.978297828703956720793870$$

(12)

```
> Em:=-1.97829782870395672079387 0;
```

$$Em := -1.97829782870395672079387\,0 \tag{13}$$

```
> evalf(sol[3]);#E0=28.87280028
```

$$29.876122069097166173032\,00 = 29.720763361662431074128\,13 \tag{14}$$

```
> E0:=29.720763361662431074128 13;
```

$$E0 := 29.876122069097166173032\,00 \tag{15}$$

```
> evalf(sol[2]);#C
```

$$C = -1.731841026798320557990073 \times 10^{-24} \tag{16}$$

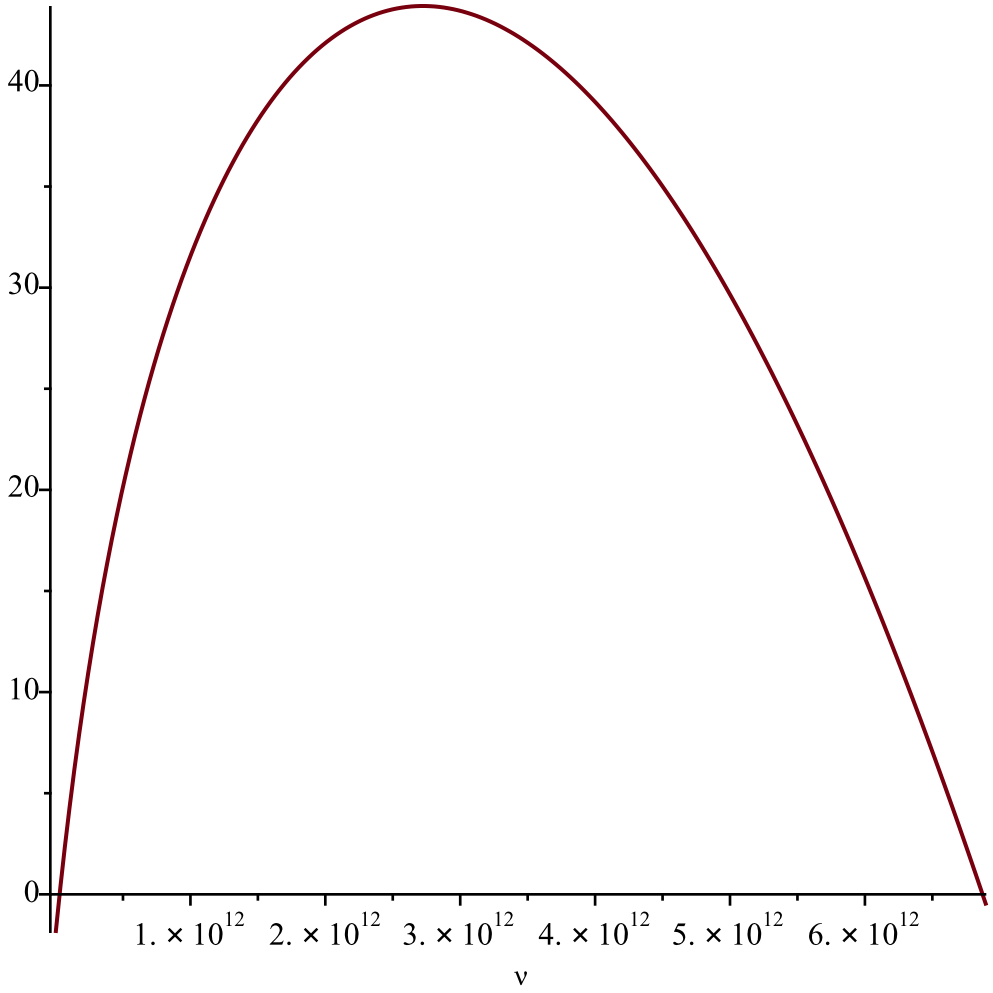
```
> C:=-1.731841026798320557990073*10^(-24);
```

$$C := -1.731841026798320557990073 \times 10^{-24} \tag{17}$$

```
> E(nu):=Em+E0*ln(1+A*2.824*10^(-18)*nu)+ C*nu^2;
```

$$E(\nu) := -1.97829782870395672079387\,0 + 29.876122069097166173032\,00 \ln\left(1 + 2.256508728000000000000000 \times 10^{-12}\,\nu\right) - 1.731841026798320557990073 \times 10^{-24}\,\nu^{2} \tag{18}$$

```
> plot(E(nu),nu=10^9..6.9*10^12);
```




```
> evalf(subs(nu=285*10^9,E(nu)));evalf(subs(nu=160.23*10^9,E(nu)));
  evalf(subs(nu=5.85*10^12,E(nu)));
```

$$12.71714922892957365625744$$

$$7.19794857385519911088324 2$$

$$18.02356542077682422671930 \quad (19)$$

```
> evalf(subs(nu=0,E(nu)));evalf(subs(nu=30.4*10^9,E(nu)));evalf
  (subs(nu=2.79*10^12,E(nu)));#43.9eV @ 15.4Ghz [30Ghz÷6.85Thz]
```

$$-1.97829782870395672079387 0$$

$$0.00230418382458687959803 9$$

$$43.91308521039839988029288 \quad (20)$$

```
> 160.23*10^9·hmw ; 285·10^9·hmw;
    #Penzias and Wilson originally detected the Cosmic Microwave
    Background 160.23÷285 GHz with an energy of ~12.64÷22.49 eV ;
```

$$1.6023000000000 \times 10^{11}\ hmw$$

$$285000000000\ hmw \quad (21)$$

```
> 160.23*10^9*hmw;#The energy of CMB
```

$$12.64291637 \quad (22)$$

```
>  http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/h21.html
> #10.9Ghz is 3.59eV on the microwave thus we can deduce the
  Planck's constatnt for miccrowave;
> E:=hmw*nu;
```

$$E := 7.890480167 \times 10^{-11}\ \nu \quad (23)$$

```
> hmw:=3.59/(10.9*10^9);
```

$$hmw := 3.293577982 \times 10^{-10} \quad (24)$$

```
> 160.23*10^9 ;#Penzias and Wilson originally detected the Cosmic
  Microwave Background 160.23GHz with an energy of ~7.1eV ;
```

$$1.602300000 \times 10^{11} \quad (25)$$

```
> c := 299792458;
  160.23*10^9;
  evalf(c/(1.9*10^11));evalf(c/(10.9*10^9));
```

$$c := 299792458$$

$$1.602300000 \times 10^{11}$$

$$0.001577855042$$

$$0.02750389523 \quad (26)$$

```
> hmw:=0.188977/(1.057*10^9);13.60569292/4;3.5904-3.401423;
```

$$hmw := 1.787861873 \times 10^{-10}$$

$$3.401423230$$

$$0.188977 \quad \textbf{(27)}$$

```
> h := 6.6260701500*10^(-34);rb:=5.239*10^(-10);
  c := 299792458;va := 343;b:=evalf(c/va)^2;
  T := 273.15 ;ec := 1.6021766340*10^(-19);Tq:= 1.765*10^(-19);
  k := 1.3806490000*10^(-23);
```

$$h := 6.626070150 \times 10^{-34}$$

$$rb := 5.239000000 \times 10^{-10}$$

$$c := 299792458$$

$$va := 343$$

$$b := 7.639292971 \times 10^{11}$$

$$T := 273.15$$

$$ec := 1.602176634 \times 10^{-19}$$

$$Tq := 1.765000000 \times 10^{-19}$$

$$k := 1.380649000 \times 10^{-23} \quad \textbf{(28)}$$

```
> 0.19/c/(-364.64^(-1)+345.35^(-1))*10^(-9);he:=h/ec;solve(0.19/c/
  (-x^(-1)+345.35^(-1))*10^(-9)=he);
```

$$4.137370747 \times 10^{-15}$$

$$he := 4.135667697 \times 10^{-15}$$

$$364.6483874 \quad \textbf{(29)}$$

> http://hyperphysics.phy-astr.gsu.edu/hbase/quantum/Lamb2.html#c1

```
> restart;# The Monopole Microawe Planck's law
```